SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

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1. Introduction:

This section introduces FreelanceShield (GigHub) — a strategic solution to the identified problem of non-payment, income instability, and lack of financial credibility among Malaysia's 1.64 million gig workers. The platform is an escrow-backed freelance marketplace where GLM-4 (ZhipuAI) is the central decision intelligence layer — not a bolt-on chatbot. It is aligned to the UMHackathon 2026 domain: AI for Economic Empowerment & Decision Intelligence.

1.1. Purpose:

- **Architecture:** A monolithic full-stack web application built on Next.js (App Router) with a Node.js/Express REST API backend. Deployed on Vercel (frontend) with PostgreSQL as the primary relational database, Redis for session and queue management, and Cloudflare R2 for encrypted file storage.
- **Data Flows:** How freelancer deliverables, brief data, GLM scoring payloads, and escrow events move through the system — from client-side encryption on the freelancer's device, through the Node.js API, into PostgreSQL and Cloudflare R2, and back to the relevant dashboard.
- **Model Process:** The end-to-end workflow of the six core features: brief validation, smart AI matchmaking, freelancer acceptance and escrow trigger, milestone management, income intelligence, and E2E encrypted delivery.
- **Role of Reference:** This document serves as the primary reference for all developers, testers, and judges to understand the high-level architecture, GLM integration design, data model, and deployment decisions behind GigHub.

1.2. Background:

Malaysia's gig economy employs 1.64 million workers contributing an estimated RM7 billion annually, yet freelancers face three structural traps that existing platforms have not resolved: non-payment after work is delivered, inability to prove income to banks for loans, and the absence of a neutral arbiter when disputes arise. The Gig Workers Act 2025 created a legal framework but did not provide enforcement tools or income analytics infrastructure.

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GigHub is a greenfield platform — there is no previous version. It is purpose-built for the Malaysian market with three architectural principles that distinguish it from existing solutions:

- **GLM is non-removable from the core loop:** Without GLM, the platform cannot validate whether a brief is enforceable, score whether a deliverable fulfils it, generate a dispute summary, power the matchmaking feed, or produce income intelligence. The escrow flow breaks at multiple points without it.

- **New capabilities introduced:** Client-side AES-256-GCM encryption before upload (files never exist in plaintext on GigHub servers), GLM-powered wage and complexity reasonableness scoring on job briefs, and GLM-generated formal PDF income statements accepted for bank loan applications — none of which exist on Fiverr, Upwork, or Malaysian alternatives.

- **Malaysia-specific compliance:** PDPA data handling, Gig Workers Act 2025 alignment, MYR-denominated escrow, and income benchmarking calibrated to Malaysian market rates by sector and location.

1.3. Target Stakeholders

Stakeholder	Role	Expectations
Freelancer	Browses job listings, accepts projects, submits deliverables, receives escrow payments, and generates income statements for bank loan applications.	Guaranteed payment before work begins via escrow. GLM-curated personalised job feed matching their skills and success history. Formal PDF income statement for bank and mortgage applications.
Company / SME	Posts structured briefs, funds escrow before work begins, reviews milestone submissions, and accesses a	Structured brief builder that prevents vague contracts. Confidence that payment is only released upon delivery. Access to

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	GLM-recommended shortlist of qualified freelancers.	pre-vetted talent matched to their specific brief.
Platform Admin	Oversees dispute escalations, reviews GLM-generated dispute summaries, issues final rulings, and monitors platform health.	Clear GLM dispute summaries with evidence trail. Role-based access to all jobs and disputes. Audit log for every escrow event.
Development Team	Builds and maintains the Next.js frontend, Node.js/Express backend, PostgreSQL schema, Redis cache, GLM integration, and Cloudflare R2 storage.	Well-documented architecture, clear API contracts, modular codebase with separation of concerns between escrow logic, GLM orchestration, and file handling.
QA Team	Validates system behaviour across all user roles, tests escrow state machine transitions, GLM output quality, and E2E encryption integrity.	Defined scope, documented request/response formats, clear acceptance criteria per feature, and edge case coverage for GLM boundary inputs.

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2. System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview:

Type	Details
System	Web Application (Mobile-responsive — React frontend, Next.js App Router)
Architecture	Monolithic full-stack with modular service layers (Escrow Engine, GLM Orchestrator, File Handler)
Deployment	Vercel (Next.js frontend + API routes), Cloudflare R2 (file storage), Supabase or Railway (PostgreSQL + Redis)
AI Layer	ZhipuAI GLM-4 — Brief Validation, Smart Matchmaking, Income Intelligence (server-side only, API key never exposed to client)

GigHub is structured as a client-server web application. The frontend is built in Next.js (App Router), serving both the freelancer dashboard and the company dashboard as mobile-responsive React pages. The backend is a Node.js/Express REST API handling escrow state machine logic, GLM prompt orchestration, file metadata extraction, and payment webhooks. All clients communicate with the backend via HTTPS REST API calls. PostgreSQL is the primary relational database for all transactional data; Redis handles session tokens, escrow race-condition locks, and GLM response caching; Cloudflare R2 stores encrypted deliverable files with a 72-hour auto-delete lifecycle.

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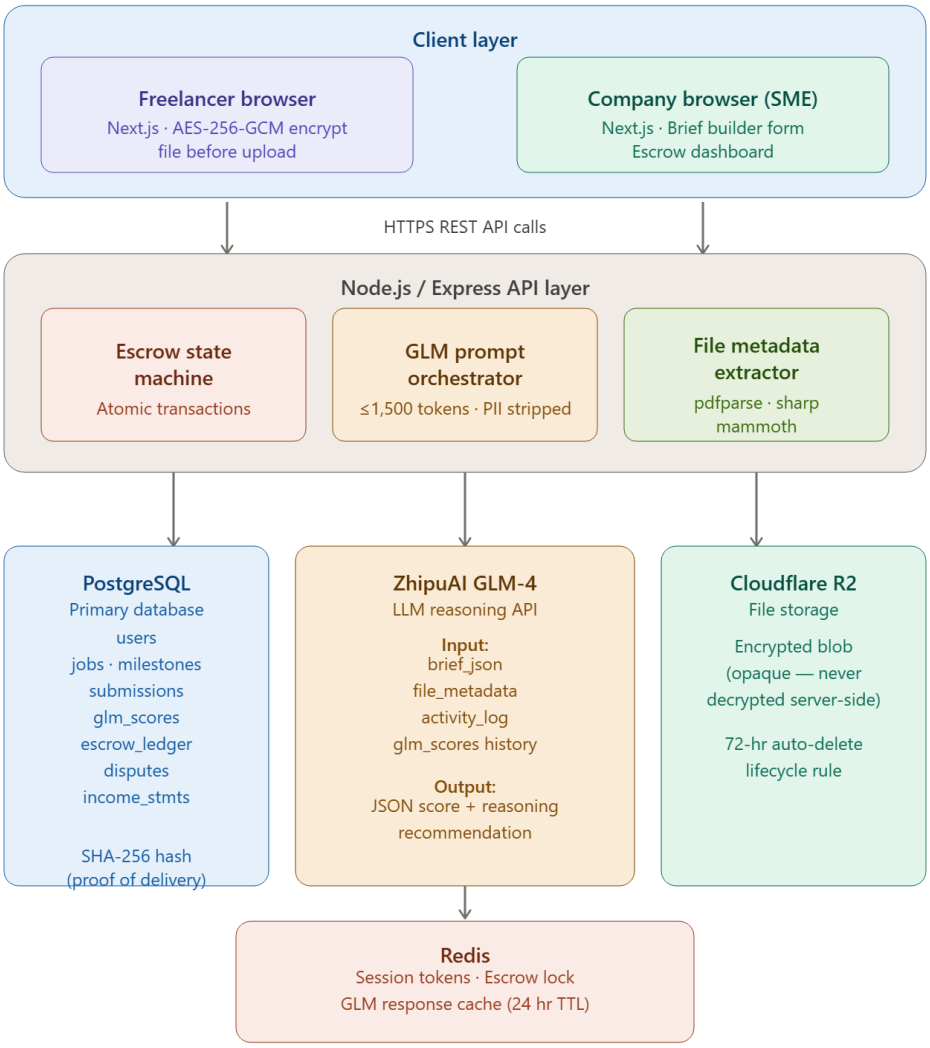
2.1.2 GLM as Service Layer

GLM-4 is integrated as a structured reasoning service — not a generic chatbot module. All GLM calls are assembled and dispatched server-side by the GLM Prompt Orchestrator. The orchestrator enforces three rules before any prompt reaches the API: (1) PII is stripped and replaced with role tokens (freelancer_1, client_1); (2) input is truncated to a maximum of 1,500 tokens to stay within cost and latency budgets; (3) GLM is instructed to return only valid JSON matching a pre-defined schema. The response is validated against a Zod schema before being persisted to the glm_scores table. GLM operates across four prompt types: brief_validation, job_matching, income_intelligence, and dispute_analysis.

2.1.3 Dependency Diagram



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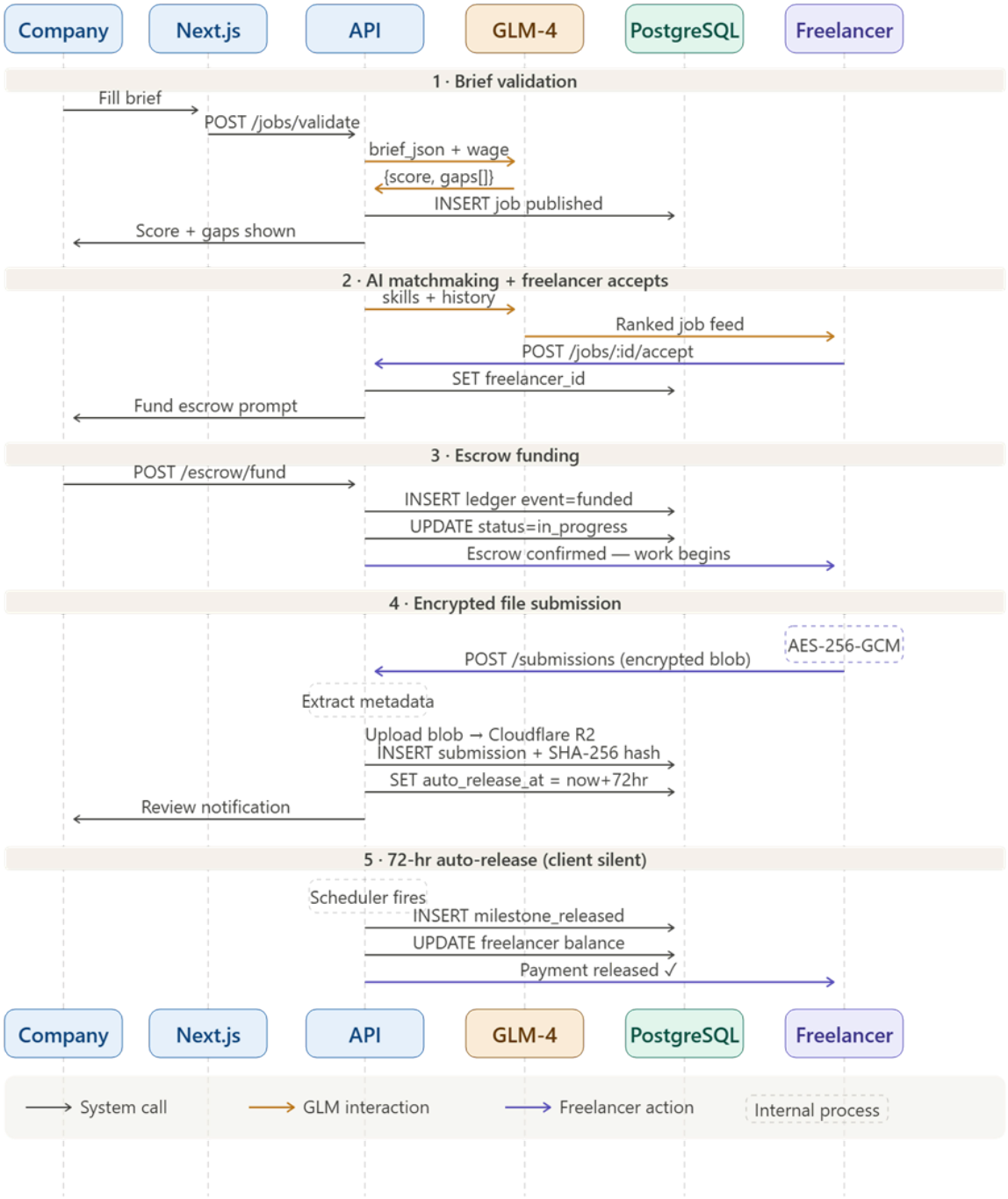
2.1.4 Sequence Diagram

The sequence below walks through the complete happy-path flow: a company posts a brief, a freelancer accepts, escrow is funded, a deliverable is submitted and auto-released after 72 hours of client non-response.



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2.2. Technological Stack

Layer	Technology	Justification
Frontend	Next.js 14 (App Router) + React	Full-stack in one repo — React frontend + API routes. Deployed free on Vercel. Mobile-responsive by default — critical as most Malaysian gig workers are mobile-first.
Backend	Node.js + Express (REST API)	Handles escrow state machine, GLM orchestration, file metadata extraction (pdfparse, sharp, mammoth), and payment webhooks. Runs as API routes within Next.js or as a standalone Express server.
Database	PostgreSQL (primary)	Relational model required — milestone state transitions must be atomic and auditable. Row-level security enforced via user_id partitioning. Escrow ledger is append-only for immutable audit trail.
Cache / Queue	Redis	Session tokens (JWT), escrow release lock (prevents double-release race condition), GLM response cache (24-hour TTL reduces API costs), real-time notification queue.
File Storage	Cloudflare R2	Zero egress fees — critical for storing large creative deliverables. 72-hour lifecycle rule for confidential file auto-deletion. Files stored as

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		encrypted blobs; SHA-256 hash persisted to PostgreSQL as proof-of-delivery.
AI / LLM	ZhipuAI GLM-4 API	OpenAI-compatible API. Handles all four prompt types: brief validation, job matching, income intelligence, dispute analysis. Free-tier credits sufficient for hackathon demo. Responses cached in Redis to minimise cost.
Payments	Stripe (sandbox) / iPay88	Stripe sandbox for hackathon demo. iPay88 for production (FPX / Malaysian online banking). PCI-DSS compliance by delegation — card data never stored on GigHub servers.
PDF Generation	pdfkit (Node.js)	Generates formal income statement PDFs from GLM income_intelligence output. Stored permanently in Cloudflare R2 (not subject to 72-hour delete rule).
Auth	JWT (HttpOnly cookies)	15-minute access tokens + 7-day refresh tokens. Role-based access control: freelancer, company, admin. Zod schema validation on all API inputs.

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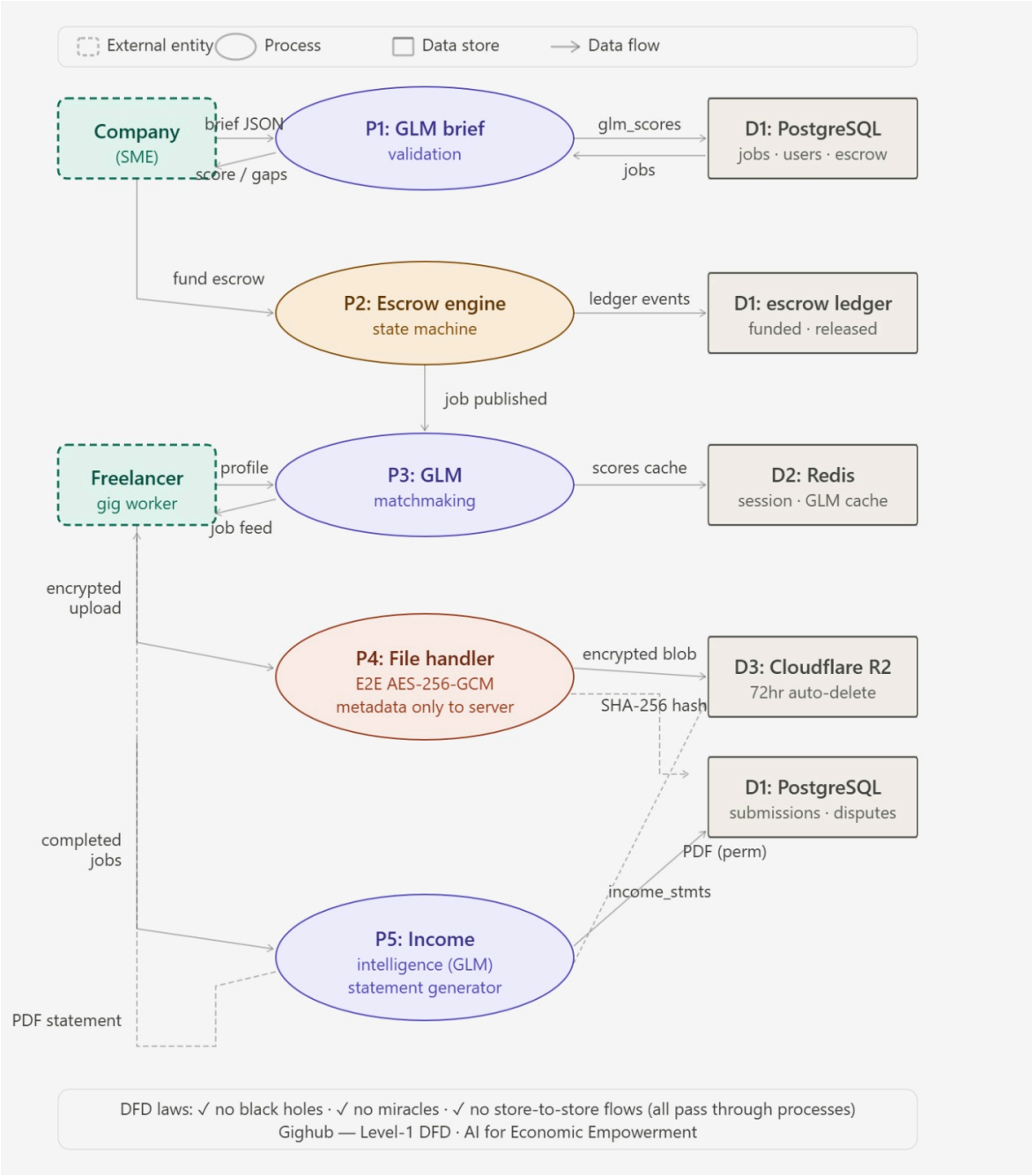
2.3. Key Data Flows

2.3.1. Data Flow Diagram (DFD)

The Level-1 DFD below illustrates data movement across the entire system, covering all external entities (Freelancer, Company, Admin), all major processes, and all data stores. DFD laws are enforced: no black holes (every process has an output), no miracles (every process has an input), and no direct data-store-to-data-store flows.



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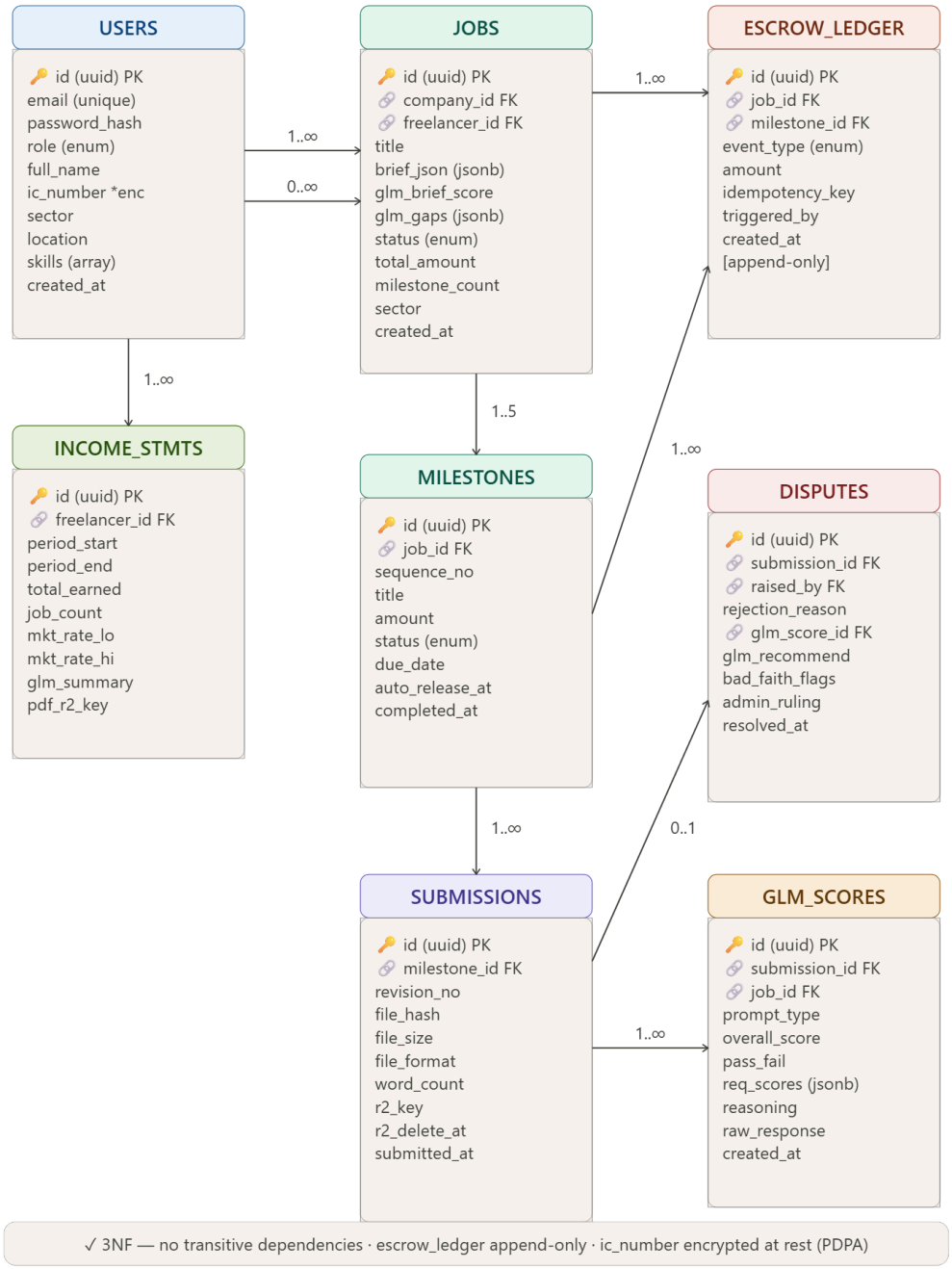
2.3.2 Normalized Database Schema

The ERD below shows the full logical data model. The schema is designed to Third Normal Form (3NF): all non-key attributes depend only on the primary key, there are no partial dependencies, and there are no transitive dependencies. The escrow_ledger table is append-only (no UPDATE operations) to guarantee an immutable financial audit trail.



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3. Functional Requirements & Scope

This section defines the core boundaries of GigHub. All six features below constitute the MVP to be built and demonstrated at UMHackathon 2026, chosen to showcase the full technical feasibility of the escrow + GLM intelligence loop.

3.1. Minimum Viable Product:

3.1.1 Core Features for Hackathon Demo

	Feature	Description
1	**Brief Builder + GLM Validation**	Companies fill a structured brief form (deliverable type, format, constraints, deadline, wage). GLM-4 scores the brief for specificity and wage/complexity reasonableness (0–100). A brief must score ≥70 to be published on the marketplace. Prevents unenforceable contracts from the start.
2	**Smart AI Matchmaking**	Two-way GLM-powered matching. Company side: when a brief goes live, GLM scans the talent pool and recommends a shortlist of qualified freelancers ranked by skill match and past submission quality (glm_scores history). Freelancer side: GLM reads the freelancer's profile, listed skills, and past GLM success rate to curate a personalised ranked job feed.
3	**Freelancer Acceptance + Escrow Trigger**	Once a brief is published, a freelancer accepts the job. The platform then prompts the company to fund the full project amount into escrow before any work begins. This sequence — accept first, then escrow — ensures no freelancer starts work without guaranteed payment, and no company is locked into escrow before a match exists.

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4	**Milestone System**	Project amount broken into up to 5 milestones. A 50/50 split option (50% upfront, 50% on delivery) is available to ease SME friction. If a client does not respond to a submission within 72 hours, the milestone is automatically released to the freelancer — preventing ghosting. Design deliverables are watermarked before preview to prevent IP theft before payment.
5	**Income Intelligence Dashboard**	Freelancer-facing dashboard showing monthly and annual earnings summaries, market rate benchmarks by skill and sector (GLM-analysed from platform data), and a GLM-generated formal PDF income statement for bank loan and mortgage applications. Directly addresses the 'unbankable' status of gig workers under the Gig Workers Act 2025.
6	**E2E Encrypted Delivery**	All deliverables are encrypted client-side (AES-256-GCM) on the freelancer's device before upload. GigHub's servers receive and store only an opaque encrypted blob — the platform never holds decryptable content. Files are hard-deleted from Cloudflare R2 after 72 hours. A SHA-256 hash is permanently recorded in PostgreSQL as immutable proof-of-delivery for dispute purposes.

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3.2. Non-Functional Requirements (NFRs)

The following system qualities are not features but are critical for reliability, security, and compliance:

Quality	Requirement	Implementation
Security	All deliverables must be encrypted end-to-end. API keys must never be exposed to the client. PDPA-compliant data handling. IC numbers encrypted at rest.	AES-256-GCM client-side encryption. GLM API key server-side only. ic_number field encrypted at rest in PostgreSQL. JWT stored in HttpOnly cookies only.
Reliability	Escrow state transitions must be atomic — a funded milestone cannot be double-released. Auto-release timer must fire reliably.	PostgreSQL atomic transactions for all escrow state changes. Redis idempotency lock on milestone release. Dead-letter queue for missed auto-release timers.
Performance	API response time < 800ms at 50 concurrent users. GLM API calls must return within 5 seconds under normal load.	Redis caches GLM brief scores for 24 hours. Zod input validation before DB write. k6 load tests enforced in CI/CD pipeline at ≥85% pass rate.
Scalability	Platform must handle concurrent brief submissions and milestone releases without race conditions.	Redis escrow lock prevents concurrent double-release. Stateless API design allows horizontal scaling on Vercel serverless functions.

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Maintainability	Codebase must be modular with clear separation between escrow logic, GLM orchestration, and file handling.	GLM Prompt Orchestrator isolated as a service module. Escrow state machine as a separate utility. File handler decoupled from submission API route.
Token Latency	GLM API calls must return within 5 seconds at p95. User must see a loading state during GLM processing.	5-second timeout on all GLM API calls. Server-Sent Events (SSE) or polling for real-time GLM score feedback on brief builder.
Cost Efficiency	Average GLM token cost per user session must stay within ZhipuAI free-tier limits for hackathon demo.	Input truncated to 1,500 tokens max before GLM call. GLM responses cached in Redis (24-hour TTL). Repeated brief validations on unchanged briefs served from cache.
Compliance	Platform must comply with Malaysia's PDPA and Gig Workers Act 2025.	Purpose-limited data collection documented in privacy policy. Users can request data deletion. Escrow transparency aligns with Gig Workers Act 2025 payment guarantee requirements.

3.3. Out of Scope / Future Enhancements

The following features are part of GigHub's grand vision but are explicitly excluded from the UMHackathon 2026 build:

- In-app messaging / real-time chat between freelancer and company
- Mobile native application (iOS / Android) — web-responsive only for hackathon
- Multi-currency support — MYR only for this build
- Third-party OAuth login (Google, LinkedIn, MyDigitalID)

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- Platform admin analytics dashboard with aggregate statistics
- Promotional offers, discount codes, or referral programs
- Scheduled milestone delivery (set a future date for automatic submission)
- Enterprise tier with client-managed storage (Google Drive, Dropbox integration)
- GLM visual quality assessment of design deliverables (glm-4v) — metadata verification used instead
- MSB licensing for real money escrow — Stripe sandbox used for hackathon; production requires Bank Negara Malaysia licensed payment partner

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4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation:

4.1.1 Grayscale Rollout & A/B Testing

For the hackathon demo, GigHub operates in a controlled demo environment with seeded test data. For post-hackathon production deployment, the following rollout strategy applies:

- **Phase 1 — Internal:** New features deployed to admin accounts only. Team validates behaviour with real data for 48 hours.
- **Phase 2 — Canary (5%):** Feature released to 5% of active freelancers and companies via feature flag in the users table. Monitoring alerts set on error rate and GLM response quality.
- **Phase 3 — Gradual rollout:** If no anomalies detected after 48 hours, rollout expanded to 25% → 50% → 100% in 24-hour increments.
- **A/B Testing:** For GLM prompt variants (e.g. brief scoring prompt v1 vs v2), both versions run in parallel for 72 hours. Versions with higher pass rate and lower dispute rate are promoted to 100%.

4.1.2 Emergency Rollback (ER) & Golden Release

Every deployment to production is tagged as a versioned release in GitHub. The last fully-passing release (all CI/CD gates green, zero P0/P1 bugs) is designated the Golden Release. In the event of a P0 or P1 incident:

- Vercel's instant rollback feature redeploys the Golden Release within 60 seconds.
- Database migrations are versioned with a down migration script — schema changes are reversible.
- Redis cache is flushed on rollback to prevent stale GLM scores from being served.
- Active users receive an in-app notification: 'We are performing maintenance. Your escrow funds are safe.'

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4.1.3 Priority Matrix

P	Level	Trigger Condition	Action
P0	Critical	Escrow double-release event detected (duplicate escrow_ledger entry)	Immediate emergency rollback to last Golden Release. Alert all active users. Manual audit of affected ledger entries.
P0	Critical	GLM API key exposed in client-side bundle or server logs	Immediate key rotation. Redeploy with corrected environment variable configuration. Security audit.
P1	High	GLM API error rate > 5% (brief validation failing for >5% of submissions)	Roll back GLM prompt version to last known-good. Enable cached score fallback. Alert team.
P1	High	Auto-release scheduler failure — milestones not releasing after 72 hours	Manual release via admin panel. Restart scheduler job. Investigate dead-letter queue.
P2	Medium	API response time > 800ms at p95 for > 10 minutes	Scale up Vercel serverless function concurrency. Investigate slow queries. Check Redis cache hit rate.
P2	Medium	R2 72-hour delete lifecycle failing (files persisting beyond window)	Manually trigger delete via R2 API. Fix lifecycle rule. Log affected file hashes.

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P3	Low	GLM brief score inconsistency (same brief returns different scores on retry)	Serve cached score. Log discrepancy. Review prompt temperature setting (set to 0 for determinism).

4.2. Monitoring:

The following monitoring plan covers system health, GLM output quality, and escrow integrity:

- **API Health:** Vercel built-in monitoring tracks response time, error rate, and function invocation count. Alert threshold: error rate > 1% over 5 minutes.
- **GLM Output Quality:** Every GLM response is validated against a Zod schema before persistence. Schema validation failure rate logged to PostgreSQL. Alert threshold: failure rate > 5% in any 1-hour window.
- **Escrow Integrity:** A scheduled job runs every 15 minutes to verify that no milestone has a released status without a corresponding escrow_ledger milestone_released event. Discrepancies trigger P0 alert.
- **Auto-Release Timer:** Dead-letter queue logs any auto-release job that fails to fire. Alert if any milestone's auto_release_at timestamp is more than 5 minutes past with status still 'submitted'.
- **R2 Lifecycle:** A daily job queries R2 for any object older than 73 hours. If found, manual deletion is triggered and a security alert is raised.
- **GLM Cost:** Daily token usage monitored against ZhipuAI account credit balance. Alert when balance drops below 20% of initial credit.

4.3. Assumptions:

The following operational and environmental conditions are assumed to be valid during development and deployment:

- Freelancers and companies have stable internet connections during brief submission, file upload, and escrow transactions — Malaysian 4G/5G penetration is sufficient for the target market.
- Freelancers access GigHub via a modern mobile browser (Chrome/Safari on Android/iOS) — no native app installation required.

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- ZhipuAI GLM-4 API free-tier credits are sufficient for hackathon demo volume (estimated < 500 GLM calls during judging).
- All monetary values are in Malaysian Ringgit (MYR) — no currency conversion is required for the hackathon build.
- Stripe sandbox webhooks are reliably delivered in the demo environment — production would require iPay88 FPX integration for Malaysian bank transfers.
- The escrow ledger balance is a PostgreSQL ledger entry, not a real bank account — production deployment requires an MSB licence or a partnership with a Bank Negara Malaysia licensed payment institution.
- File deliverables are assumed to be standard creative formats: PDF, DOCX, PNG, JPG, ZIP — no proprietary or executable file types.

4.4. External Dependencies:

Tool / Service	Purpose	Risk & Mitigation
ZhipuAI GLM-4 API	Core LLM reasoning engine — handles brief validation scoring, job matchmaking, income intelligence analysis, and structured JSON output generation.	High — if GLM API is rate-limited or down, brief validation, matchmaking, and income statements fail. Mitigation: Redis cache for repeated brief scores (24hr TTL); graceful degradation UI state shown to users during outage; fallback to manual admin review for disputes.
Cloudflare R2	Encrypted deliverable file storage. 72-hour auto-delete lifecycle for confidential files. Permanent storage for income statement PDFs.	Medium — if R2 is unavailable, file uploads fail. Mitigation: retry queue in Redis; user notified to retry upload; submission not marked complete until R2 confirms storage.
Stripe (sandbox) / iPay88	Payment processing for escrow funding. PCI-DSS compliance by delegation. Webhook events	High — if the payment provider is down, escrow cannot be funded and projects cannot start. Mitigation:

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	trigger escrow_ledger funded event.	webhook retry logic; idempotency keys on all payment events; clear user-facing error state.
PostgreSQL (Supabase/Railway)	Primary relational database for all transactional data. Escrow ledger, GLM scores, user data, milestone state.	Critical — database unavailability stops all platform operations. Mitigation: Supabase/Railway managed hosting with automatic backups; connection pooling via pgBouncer.
Redis (Upstash)	Session tokens, escrow race-condition lock, GLM response cache, notification queue.	Medium — if Redis is unavailable, session management and escrow locking degrade. Mitigation: fall back to stateless JWT validation; disable concurrent release protection (log warning); Upstash serverless Redis has 99.99% SLA.
Vercel	Frontend deployment and API route hosting (Next.js serverless functions).	Medium — Vercel outage affects all platform access. Mitigation: Vercel has 99.99% uptime SLA; no single-region dependency for serverless functions.

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5. Project Management & Team Contributions

The UMHackathon 2026 preliminary round spans approximately 10 days. The timeline below follows a two-week sprint structure, with parallel tracks for backend development, GLM integration, frontend build, and documentation.

5.1. Project Timeline:

Period	Phase	Key Deliverables
Day 1–2	Project Planning & Architecture	Finalise feature scope, confirm tech stack, set up GitHub repo, initialise Next.js project, design PostgreSQL schema, configure Supabase/Railway database, set up Cloudflare R2 bucket and Stripe sandbox account.
Day 3–4	Core Backend — Auth, Jobs & Escrow	Implement JWT authentication (register/login). Build job CRUD API with Zod validation. Implement escrow state machine (funded → in_progress → released). Set up escrow_ledger append-only table. Integrate Stripe webhook for escrow funding event.
Day 5	GLM Integration —Brief Validation & Matchmaking	Build GLM Prompt Orchestrator service module. Implement brief_validation prompt (score + gaps + clarifying questions). Implement job_matching prompt (freelancer feed + company shortlist). Integrate Redis caching for GLM responses (24hr TTL). Test with sample briefs.
Day 6	Milestone System & File Handling	Implement milestone API (create, submit, approve, auto-release). Build file metadata extractor (pdfparse, sharp, mammoth). Implement client-side AES-256-GCM

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		encryption flow. Integrate Cloudflare R2 upload with 72-hour lifecycle rule. SHA-256 hash persistence.
Day 7	Income Intelligence & PDF Generation	Implement income_intelligence GLM prompt. Build income_statements table and API. Integrate pdfkit for formal PDF income statement generation. Implement freelancer dashboard earnings summary and market rate benchmark display.
Day 8	Frontend—All Dashboards	Build company dashboard (brief builder, escrow tracker, active jobs, freelancer shortlist). Build freelancer dashboard (job feed, milestone tracker, income dashboard, PDF download). Build admin dispute panel. Mobile-responsive styling throughout.
Day 9	Testing, Bug Fixes & CI/CD	Write Jest unit tests for escrow state machine and GLM prompt builder. Write Supertest integration tests for full escrow flow. Set up GitHub Actions CI pipeline. Run k6 load test for API performance. Fix identified bugs. Validate all 6 MVP features end-to-end.
Day 10	Demo Preparation, Documentation & Submission	Finalise demo script (escrow flow + GLM brief scoring demo). Record demo video. Complete SAD and QATD documentation. Deploy to Vercel production URL. Submit all deliverables by deadline.

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5.2. Team Members Role:

Thong Poh Yoke — Fullstack Developer (Lead)	
Responsibilities	Next.js frontend architecture and React component development. Node.js/Express API design and implementation. PostgreSQL schema design and migrations. Escrow state machine logic. Stripe/iPay88 webhook integration. GitHub repository setup and CI/CD pipeline configuration (GitHub Actions).

Leow Wei Xin — Fullstack Developer	
Responsibilities	GLM Prompt Orchestrator service module — all four prompt types (brief validation, matchmaking, income intelligence, dispute). Redis integration (session management, escrow lock, GLM cache). Cloudflare R2 file storage and lifecycle rules. AES-256-GCM client-side encryption flow. File metadata extraction (pdfparse, sharp, mammoth). pdfkit income statement PDF generation.

Richie Wong Yu Zhi — Documentation & Pitch Lead	
Responsibilities	System Analysis Documentation (SAD). Quality Assurance Testing Documentation (QATD). Product pitch deck creation. Solution brief and narrative writing. Hackathon submission writeup. Feature scope alignment and documentation of all architectural decisions.

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Chan Jin Xuanwhat — Video Editor & UI/UX Designer	
Responsibilities	Demo video recording, editing, and production. UI/UX design for all dashboard screens (Figma wireframes). Brand identity for GigHub (logo, colour palette, typography). Frontend visual polish and mobile-responsive design review. Presentation slide design for pitch deck.

5.3. Recommendations:

Recommendation	Rationale & Implementation
Redis GLM Response Cache	Cache GLM brief scores with a 24-hour TTL key based on a hash of the brief_json. This prevents redundant API calls when a company resubmits an unchanged brief, reducing ZhipuAI credit consumption by an estimated 40–60% during peak demo usage.
PostgreSQL Connection Pooling	Use PgBouncer (built into Supabase) or a connection pool of max 10 connections in the Node.js API. Vercel serverless functions can spawn many parallel instances — without pooling, each cold start creates a new DB connection and exhausts the PostgreSQL connection limit under load.
Zod Input Validation as Middleware	Apply Zod schema validation as Express middleware on every POST/PATCH route before any database write. This prevents malformed data from reaching PostgreSQL and produces structured 422 error responses that the GLM output test cases can assert against.

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Idempotency Keys on All Payment Events	Generate a UUID idempotency_key for every escrow operation and store it in the escrow_ledger table with a unique constraint. This ensures Stripe webhook retries and concurrent release attempts cannot create duplicate ledger entries — the most critical reliability risk in the system.
GLM Prompt Temperature = 0	Set temperature: 0 in all GLM API calls. This makes GLM responses deterministic — the same brief will always return the same score, making the Redis cache effective and making AI-01 to AI-04 test assertions reliable across runs.
Watermark Generation Server-Side	Generate watermarked design previews on the server (using sharp) rather than client-side. This ensures the watermark cannot be bypassed by a client-side script. Store the watermarked preview separately in R2 as a non-encrypted public preview URL, while the encrypted original is only released after escrow payment.

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